MATLAB Scripts:

1. **File format input:**
   a. **Excel (.xlsx):** Geometry for simulations and experiments. Trajectories from experiments
   b. **Mat (.mat):** Simulation data and data analysis from experimental or simulation data.
2. **Open MATLAB script a1_readdata.m, the script to read the Excel file (.xlsx) for each loaded pillar geometry**
   a. Change value on **line 20**: flag_data=n; where "n" represents the type of data to load (Sim = simulations, Exp = Experiment, AML = favorable, CML = unfavorable):

      1: Sim AML

      2: Exp AML

      3: Sim CML

      4: Exp CML

   b. Run the script (hit the upper green triangle). The command window will show the name of the file to load. A window will show up asking for the Excel geometry file. Load one of the following files, respectively:

   Favorable Conditions

   Simulation:
   Sim_PM_Geometry-AML_1µm_FavCond-(IS_2mM-pH_4)_porosity_37
   Experiment:
   Exp_PM_Geometry-AML_1µm_FavCond-(IS_2mM-pH_4)_porosity_37

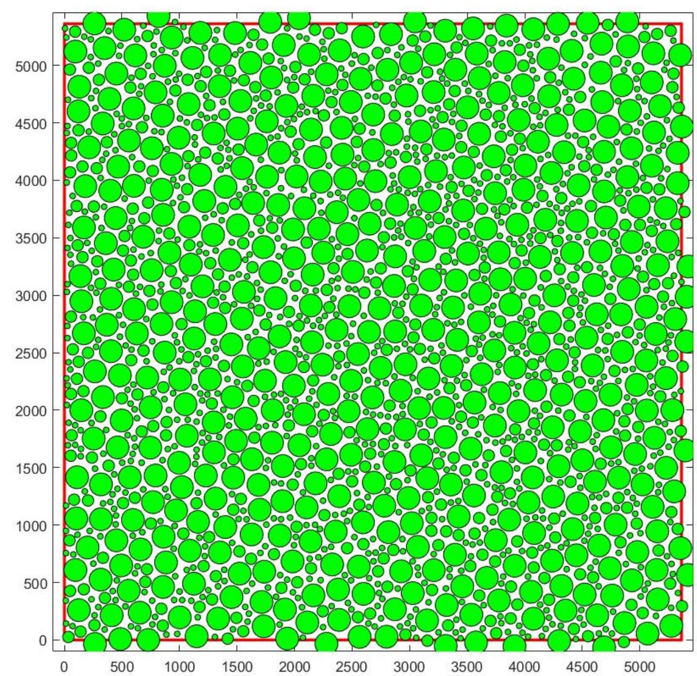
   Unfavorable Conditions

   Simulation:
   Sim_PM_Geometry-CML(0.95µm)_UnfavCond(IS_6mM-pH_6.5) _porosity_37
   Experiment:
   Exp_PM_Geometry-CML(0.95µm)_UnfavCond(IS_6mM-pH_6.5) _porosity_37

   c. The porous media geometry will be saved as an array of pillar centers and radii (x, y, r) in mat. file:
      1. Sim_geo_AML.mat
      2. Exp_geo_AML.mat
      3. Sim_geo_CML.mat
      4. Exp_geo_CML.mat

3. **Open MATLAB script a2_subset_geo.m: Script to read _geo_.mat file to create a 2D pillar image.**
    a. Change value on **line 13**: flag_data=n; where "n" represents the geometry mat. file to load.
    b. Run the script (hit the upper green triangle).
    c. The subset file will save a mat. file based on field of view specified in input Excel file:
        1. Sim_sub_geo_AML
        2. Exp_sub_geo_AML
        3. Sim_sub_geo_CML
        4. Exp_sub_geo_CML

    d. The script also generates a 2D plot within the field of view:



4. **Open MATLAB script a3_a_sepdist_vel.m, the script creates a condensed version (reducing the number colloid translation after it gets attached) of the simulation data, and the array will be save in New_Array3.**
    a. Change value on **line 10**: flag_data=n; where "n" represents the type of data to load from **simulations** to load:
        1. Sim AML: Fav_100nm_complete_Array.mat
        3. Sim CML: Unfav_100nm_complete_Array.mat

b. Change the name of the array (**line 85**) or file (**line 86**)
c. Run the script (hit the upper green triangle).
d. The condensed arrays will be saved as a mat. file:
   1. Sim-New_Array3_Array_AML.mat
   3. Sim-New_Array3_Array_CML.mat

e. The condensed arrays and number of translations for each trajectory will be saved a mat. file:
   1. Sim-New_Array3_File_AML.mat
   3. Sim-New_Array3_File_CML.mat

5. **Open MATLAB script a3_b_sepdist_vel.m**

   With the Exp prompt, the script will use experiment Excel files to produce a 3D trajectory matrix (Trajs(ic,1:11,i)) containing for each translation (ic) in each colloid trajectory (i) and (1:11) containing the frame #, x, y, time, sep. dist., and inst. vel.. Additionally, a Trajsadj(ic,1:4,i) is generated with each each translation (ic) in each colloid trajectory (i) and (1:4) containing corrected sep. dist., adjusted vel. to set minimum to zero (not negative), norm. corrected sep. dist., norm. adjusted vel.

   a. Change value on **line 10**: flag_data=n; where "n" represents the type of data to load from **experiments**, as follows:
      2: Exp AML
      4: Exp CML
   b. Check the output file name in **line 243**
   c. Run the script (hit the upper green triangle). A window will show up asking for the Excel file that contains trajectory information. Click on one of the following files:
      2: Exp_AML_trajectories
      4: Exp_CML_trajectories
   d. Run the script (hit the upper green triangle), and the Trajs and Trajsadj arrays will be saved as a mat. file:

      2: Exp-SepDist_Velocity-complete-AML
      4: Exp-SepDist_Velocity-complete-CML

   With the Sim prompt, the script will use simulation .mat files to produce a 3D trajectory matrix (Trajs(ic,1:11,i)) and Trajsadj(ic,1:4,i) as described above.

e. Change value on **line 10**: flag_data=n; where "n" represents the type of data to load from **simulations**, as follows:
   1: Sim AML
   3: Sim CML
f. Double check the loaded Sim mat. file in **line 59**
   1: Sim-New_Array3_File_AML.mat
   3: Sim-New_Array3_File _CML.mat
g. Check the output file name in **line 243**
h. Run the script (hit the upper green triangle), and the Trajs and Trajsadj arrays will be saved as a mat. file:

   1: Sim-SepDist_Velocity-all-complete-AML

   3: Sim-SepDist_Velocity-all-complete-CML

6. **Open MATLAB script a4_intercepts.m, this script uses arrays Trajs and Trajsadj to determine portions of trajectory meeting interception criteria to produce the arrays (intercept(ic,1:6,i,j), x_int(ic,i,j), time_int(ic,i,j), int_order(ic,i,j), y_int(ic,i,j), velnorm_int(ic,i,j), sepdist_int(ic,i,j), where j represents interception criterion used: 1 = sep. dist. < 100 nm, 2 = threshold norm. vel. + sep. dist. < 7 µm.**

   With the Exp prompt, the script will use simulation .mat files to produce a 3D trajectory matrix (intercept(ic,1:11,i)) as described above.

   a. Change value on **line 11**: flag_data=n; where "n" represents the type of data to load from **experiments (recall must be run before simulations)**, as follows:
      2: Exp AML
      4: Exp CML
      The experimental data (attachment x-coordinates) will be used to compare and generate the files from simulation database as explained in Step 6.

   b. If you want to explore the effect of different separation distance criterion, you can change them in **line 39** and the normalized velocity criteria in **line 40**.
   c. Change the name of the output file in **line 389**
   d. Run the script (hit the upper green triangle). The file will be saved as a mat. file, e.g.:

      2. Exp-AML-Sepdist_7.00µm-Normvel_0.020-all_Complete.mat

      4. Exp-CML-Sepdist_7.00µm-Normvel_0.020-all_Complete.mat

With the Sim prompt, the script will use simulation .mat files to produce a 3D trajectory matrix (intercept(ic,1:11,i)) as described above.

e. Change value on **line 11**: flag_data=n; where "n" represents the type of data to load from **experiments (recall must be run before simulations)**, as follows:
   1: Sim AML
   3: Sim CML
f. The separation distance criterion of **100 nm** is **predefined in line 43** (normalized velocity threshold not apply but it is defined in line 44), **do not change it**. If you want to explore the effect of different separation distance criterion, you can change them in **line 46** and the normalized velocity criteria in **line 47**.
g. Change the name of the output file in **lines 383 and 386,** for 100 nm separation distance criterion and normalized velocity (with 7µm separation distance), respectively.
h. Run the script (hit the upper green triangle). The files will be saved as a mat. file, e.g.:
   1. Sim-AML-Sepdist_0.10µm-Normvel_1.000-all_Complete.mat
      Sim-AML-Sepdist_7.00µm-Normvel_0.020-all_Complete.mat
   3. Sim-CML-Sepdist_0.10µm-Normvel_1.000-all_Complete.mat
      Sim-CML-Sepdist_7.00µm-Normvel_0.020-all_Complete.mat

7. **Open MATLAB script a5_data_extraction_Sim_100nm.m, this script will use the experimental x-coordinates of attachment to find the best matches in attachment location for favorable and unfavorable conditions.** There is no need to select any file (all of them are uploaded in the same folder).
   a. Change the name to **AML** or **CML line 8** to refer to favorable and unfavorable conditions from **experiments.**
      i. If you need to increase or decrease the number of extracted trajectories (defined 214 trajectories).
         1. Go to **line 23** to modify the **lower boundary of the field of view** (set for experiments FOV).
         2. Go to **line 33** and **line 25** to modify the **tolerance** under **favorable** (AML) and **unfavorable** conditions, respectively.
            a. In **line 25**, the value was set to **inf** to get **all trajectories**. It can be changed to 37 µm to get 214 trajectories**,** to get all trajectories within the field of view under **favorable conditions**.
            b. In **line 33**, the value was set to **inf** to get **all trajectories**. It can be changed to 20 µm to get 214 trajectories, to get all

trajectories within the field of view under **unfavorable conditions**.

b. Run the script (hit the upper green triangle).
    i. A file with simulation attachment locations related to experiments will be saved as a .mat file:
        1. Favorable: Fav_100nm_ complete_Array.mat (see name in **line 153**)
        2. Unfavorable: Unfav_100nm_complete_Array.mat (see name in **line 155**)

8. **Refer again to step 4 and 5 to obtain simulation files that contain arrays: Trajs, Trajsadj, and intercept.**
    a. **Step 4 (a3_c_sepdist_vel.m):** Refer from steps d to h.
    b. **Step 5 (a4_intercepts.m):** Refer to steps c, and d.
    c. Run the script (hit the upper green triangle).

9. **Open MATLAB a5_plots.m: Script for plotting: (1) colloid trajectories, absolute sep. dist, and norm. vel. (2) number of translations discriminated by criteria vs distance, and vs time. (3) cumulative number of translations, interception order vs interception mean-distance (avg[min & max]), and interception difference (relative to 100 nm sep. dist.) vs maximum x-distance (attachment location)**

    This script is used to evaluate the behavior of a single trajectory (flag_plot_1) or from a subset of trajectories (flag_plot_2 and flag_plot_3).

    a. Change value on **line 10**: flag_data=n; where "n" represents the conditions data to load (Sim, Exp, AML or CML)

    b. Change value on **line 17** (flag_plot_1), **line 18** (flag_plot_2), and **line 19** (flag_plot_3) as 1 for plotting as 0 for not plotting.
        i. **flag_plot_1** = **figure (1):** trajectory (1), vel nom. (2) and sep dist. (3) vs x-distance (µm)
        ii. **flag_plot_2 = figure (2):** Colloid trajectory (1), vel_norm/sep_dist vs distance (2), vel_norm/sep_dist vs time (3)
        iii. **flag_plot_3 = figure(3):** Translation number (1), Interception order (2) vs mean x-distance (µm). And for simulations only (3) interception difference relative to 100 nm vs maximum x-distance (attachment location).
    c. Select the trajectory to be plotted in **line 25** (ini_traj) and **line 26** (end_traj):

   i. Check the total number of trajectories by using **disp(npart)** in the command window
   ii. **flag_plot_1** = **figure (1)**, there should be a single trajectory (ini_traj=end_traj)
   iii. **flag_plot_2 = figure (2)**, there can be multiple trajectories where ini_traj<end_traj (e.g. ini_traj=1, end_traj=31)
   iv. **flag_plot_3 = figure(3)**, there can be multiple trajectories where ini_traj<end_traj (e.g. ini_traj=1, end_traj=31)

  d. **Only for Figure (1)** the criteria to be analyzed:
   i. Select the separation distance criterion in **line 30 (j_sep_crit_idx)**
      1. Set 1 for Sim (sep. dist=100nm) or Exp (sep. dist=7µm).
      2. Set 2 for Sim (sep. dist=7µm)
   ii. Select the normalized velocity threshold criterion in **line 32 (j_thresh_idx).**
      1. Set 1 normalized velocity threshold criterion of 1 for Sim with a separation distance of 100 nm.
      2. Set 1,2,3 normalized velocity threshold criterion [0.015 0.020 0.025] for Sim and Exp, both with a separation distance of 7 µm.

  e. Run the script (hit the upper green triangle).

10. Open MATLAB a6_xyt_numint_data_excel.m, this script is used to extract the data of x,y attachment location, interception order and elapsed time for each trajectory under favorable and unfavorable conditions as a single Excel file. This data will be used to create interception order vs distance histograms.

  a. Change value on **line 12 and line 13**, for separation distance and normalize velocity thresholds, respectively. The first argument in the matrix will correspond to the simulation conditions, and the second to the experiment.
   i. **line 12:** sep_dist = [0.1, 7] = [Sim Exp] (µm)
   ii. **line 13:** thresh_norm_vel = [1, 0.02] = [Sim Exp] (-)

  b. Make sure that load **file_name** in **line 23, 49, 80, and 107** is the same.

  c. The matrices will be saved in different sheets as output data in an Excel file with the name defined in **line 133** (e.g., output_data-Sim_Exp-histogram(numint_vs_x-coordinate)_7µm_0.02_normvel.xlsx)

  d. Run the script (hit the upper green triangle).

11. Open MATLAB a6_vel_data_excel.m, this script for extracting velocity data and t-test (p-values) in an Excel spreadsheet file

  a. The matrices will be saved in different sheets as output data in an Excel file with the name defined in **line 11** (e.g., output_data-Sim_Exp-velocity_analysis.xlsx).

b. Check the loaded criteria (**sep_criterion:** separation distance and **thresh:** normalized velocity threshold) from **line 22** to **34.**
    i. **Line 24:** conditions for Sim-AML
    ii. **Line 27:** conditions for Exp-AML
    iii. **Line 30:** conditions for Sim-CML
    iv. **Line 33:** conditions for Exp-CML
c. Make sure that **file_name** in **line 37** is the same output mat. file from script **a4_intercepts.m**.
d. Run the script (hit the upper green triangle).

12. Open MATLAB a6_xyt_numint_data.m, this script for extracting: numint data.

a. The names of the output file will be save with the name defined in **line 132.**
b. The separation distance criteria are defined i**n line 12** for sep. dist. of 100 nm and 7 µm, respectively.
c. The normalized velocity thresholds criteria are i**n line 13** for sep. dist. of 100 nm and 7 µm, respectively.
d. Make sure that the same in **lines 23, 49, 80 and 107** are the same output mat. file from script **a4_intercepts.m**.
e. Run the script (hit the upper green triangle).
f. Output file will be created.
    i. output_all_data_Sim_Exp-histogram(numint_vs_x-coord)_7µm_0.02_normvel.xlsx)

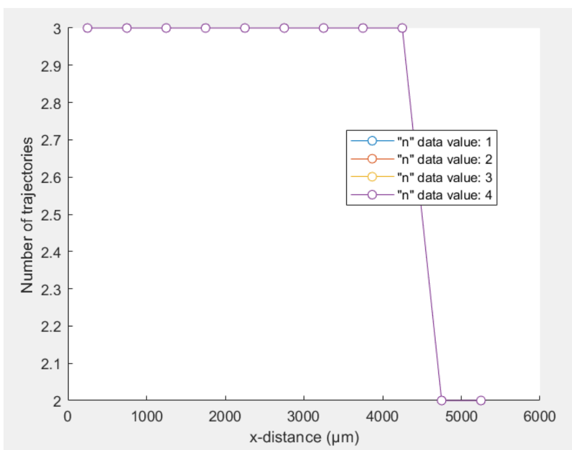
13. Open MATLAB a7_min_max_x_t_ntrans_data.m, this script for extracting: x_int_minimum_x, x_int_maximum_x, x_int_minimum_t, x_int_maximum_t, ntrans_int data.

a. The names of the output files (from arrays defined in **line 17**) are defined in **lines 10-14.**
b. The separation distance criteria are defined i**n line 28, 33, 38, and 43** for Sim-AML, Exp-AML, Sim-CML and Exp-CML, respectively.
c. The normalized velocity thresholds criteria are i**n line 48 and 50** for sep. dist. of 100 nm and 7 µm, respectively.
d. Make sure that the same in **line 54** is the same output mat. file from script **a4_intercepts.m**.
e. Run the script (hit the upper green triangle).
f. Output files will be created.
    i. output_all_x_int_min_x.xlsx

ii. output_all_x_int_max_x.xlsx
iii. output_all_x_int_min_t.xlsx
iv. output_all_x_int_max_t.xlsx
v. output_all_ntrans_int.xlsx

Creation of subset based on AML, CML, Sim and Exp data

1. **Open MATLAB b1_rows_2_extract.m, this script will load and create subsets based on mat. files created in a4_intercepts.m**
    a. Make sure that the same in **line 34** is the same output mat. file from script **a4_intercepts.m**.
    b. The trajectories will be classified according to a predefined number of bins (11), calculated in **line 71**.
        i. In **lines 84 – 87**, the criteria to define the number of interceptions will be calculated for each bin. Predefined as 3 for the first 9 bins, and as 2 for the remaining bins.
    c. Output mat. file name is defined in **line 155.**
    d. Run the script (hit the upper green triangle).
    e. The generated plot will show the number of trajectories for each subset. The "n" data value represents the type of data to load (**See step 2**)



    f. The mat. file will be saved as:
        i. Output file: Sim_Exp_subset_new_arrays.mat
2. **Open MATLAB b2_data_ext_subset.m, this script to extract subset data from simulations (100 nm sep dist) and experiments indices extracted in b1_rows_2_extract.m script**
    a. Make sure that the same in **line 11** is the same output mat. file from script **b1_rows_2_extract.m**.
    b. Check the names of mat. files to load in **lines 22 (Sim-AML), 31 (Sim-CML)**, which should be the same as the output array file from script a3_a_sepdist_vel.m
        i. 1. Sim-New_Array3_Array_AML.mat
        ii. 3. Sim-New_Array3_Array_CML.mat

    c. Check the names of excel file in **lines 41 (Exp-AML), and 91 (Exp-CML).**
        i. Exp_AML_trajectories.xlsx

    ii. Exp_CML_trajectories.xlsx

  d. Check the names of output mat. files to save in **lines 26 (Sim-AML), 35 (Sim-CML),** and excel file in **81 (Exp-AML), and 131 (Exp-CML).**

  e. Run the script (hit the upper green triangle).

    i. mat. files for simulation arrays (100 nm).

    ii. xlsx. files for experiment excel files.

3. The subset analysis should be addressed by following the same process as for the complete dataset (Steps: 4-6 and 9-13**). Make sure of changing the name of the output files in every script, to avoid overwriting.**

  a. Step 1-3: Skip these steps since geometry and sub geometry .mat files are already created.

<u>Step 5 can be addressed instead of Step 4</u>

  b. Step 4 (a3_a_sepdist_vel.m**):**

    i. Change value on **line 10**: flag_data=n; where "n" represents the type of data to load from **simulations** (**1. Sim AML and 3. Sim CML).**

    ii. Change the name of the array (**line 85**) or file (**line 86**) including the word **subset** to avoid overwriting.

    1: Sim-New_Array3_File_AML_subset.mat
    3: Sim-New_Array3_File_CML_subset.mat

    iii. Run the script (hit the upper green triangle).

    (a3_b_sepdist_vel.m**):**

    iv. Change value on **line 10**: flag_data=n; where "n" represents the type of data to load from **simulations and experiments**.

    v. Double check the loaded Sim mat. file in **line 59**
    1: Sim-New_Array3_File_AML_subset.mat
    3: Sim-New_Array3_File_CML_subset.mat

    vi. Change the output file name in **line 243** including the word **subset** to avoid overwriting.

    vii. Run the script (hit the upper green triangle).

  c. Step 5 (a3_c_sepdist_vel_subset.m**)**

    i. Change value on **line 10**: flag_data=n; where "n" represents the type of data to load from simulations and experiments.

    ii. Change the output file name in **line 287** including the word **subset** to avoid overwriting.

d. **Step 6 (**a4_intercepts.m**)**
   i. Change value on **line 11**: flag_data=n; where "n" represents the type of data to load from **simulations and experiments**.
   vi. For experiments change the name of the output file in **line 389** including the word **subset** to avoid overwriting.

      Exp-AML-Sepdist_7.00µm-Normvel_0.020-all_Complete_subset.mat

      Exp-CML-Sepdist_7.00µm-Normvel_0.020-all_Complete_subset.mat

   vii. For simulations, **c**hange the name of the output file in **lines 383 and 386** including the word **subset** to avoid overwriting.

      Sim-AML-Sepdist_0.10µm-Normvel_1.000-all_Complete_subset.mat

      Sim-AML-Sepdist_7.00µm-Normvel_0.020-all_Complete_subset.mat

   viii. Run the script (hit the upper green triangle).

e. **Step 9 (**a5_plots.m**)**
   i. Change value on **line 10**: flag_data=n; where "n" represents the conditions data to load (Sim, Exp, AML or CML).
   ii. Change value on **line 17** (flag_plot_1), **line 18** (flag_plot_2), and **line 19** (flag_plot_3) as 1 for plotting as 0 for not plotting.
   iii. Select the trajectory to be plotted in **line 25** (ini_traj) and **line 26** (end_traj):
   iv. **Only for Figure (1)** the criteria to be analyzed:
      1. Select the separation distance criterion in **line 30 (j_sep_crit_idx)**
      2. Select the normalized velocity threshold criterion in **line 32 (j_thresh_idx).**
   v. Run the script (hit the upper green triangle).

f. **Step 10 (**a6_xyt_numint_data_excel.m**)**
   i. Change value on **line 12 and line 13**, for separation distance and normalize velocity thresholds, respectively.
   ii. Make sure that load **file_name** in **line 23, 49, 80, and 107** is the same and referred to the subset files.

iii. The matrices will be saved in different sheets as output data in an Excel file with the name defined in **line 133**, include the word **subset** to avoid overwriting.
iv. Run the script (hit the upper green triangle).

g. Step 11 (a6_vel_data_excel.m**)**

i. The matrices will be saved in different sheets as output data in an Excel file with the name defined in **line 11** (e.g., output_data-Sim_Exp-velocity_analysis.xlsx), include the word **subset** to avoid overwriting.
ii. Check the loaded criteria (**sep_criterion:** separation distance and **thresh:** normalized velocity threshold) from **line 22** to **34.**
iii. Make sure that **file_name (including the word subset)** in **line 37** is the same output mat. file from script **a4_intercepts.m**.
iv. Run the script (hit the upper green triangle).

h. Step 12 (a6_xyt_numint_data.m**)**

i. The names of the output file will be save with the name defined in **line 132,** include the word **subset** to avoid overwriting.
ii. The separation distance criteria are defined i**n line 12** for sep. dist. of 100 nm and 7 µm, respectively.
iii. The normalized velocity thresholds criteria are i**n line 13** for sep. dist. of 100 nm and 7 µm, respectively.
iv. Make sure that the same in **lines 23, 49, 80 and 107** are the same output mat. file from script **a4_intercepts.m**.
v. Run the script (hit the upper green triangle).

i. Step 13 (a7_min_max_x_t_ntrans_data.m**)**

i. The names of the output files (from arrays defined in **line 17**) are defined in **lines 10-14,** include the word **subset** to avoid overwriting.
ii. The separation distance criteria are defined i**n line 28, 33, 38, and 43** for Sim-AML, Exp-AML, Sim-CML and Exp-CML, respectively.
iii. The normalized velocity thresholds criteria are i**n line 48 and 50** for sep. dist. of 100 nm and 7 µm, respectively.
iv. Make sure that the same in **line 54** is the same output mat. file from script **a4_intercepts.m**.
v. Run the script (hit the upper green triangle).